

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

2. RF and M/W Transmission Lines

6 hrs · asked in **all 24 papers** · **297 marks, 14.3% of the archive** · **252 numerical (85%) + 45 theory**, 12.4 per paper · 4 syllabus sub-topics → the Smith chart chapter: almost every mark here is a graphical construction.

READ THIS FIRST

This is the second-heaviest chapter in the subject and the most predictable one. Of its 297 marks, roughly 230 are stub-matching constructions, and **double-stub tuning alone appears in 17 of the 24 papers** — the single most reliably asked item in RF and Microwave Engineering. Learn the construction, not the answers.

1. **Double-stub matching** (17 papers, usually 8–12 marks) → master one procedure: spacing circle, constant- g arc, rotate, cancel. It works for every variant the examiner prints (stub spacing $\lambda/8$ or $3\lambda/8$, short or open, first stub at the load or offset).
2. **The S-matrix rider** (10 papers) → “prepare the S-matrix of the matched network” is tacked onto most stub questions and is worth 2–6 marks for two lines of work. See §2.8; do not leave it.
3. **Single-stub matching** (6 papers) and **the immittance chart** (5 papers) → both are short, self-contained and frequently the easy half of a long question.

Numerical work is in the companion, *Ch2 – RF and MW Transmission Lines – Numerical Problems & Solutions*, where all 24 PYQ constructions are solved step by step.

2.1 Types of Transmission Lines

Differentiate the behaviour of a transmission line at conventional low frequency and at microwave bands. Compare microwave transmission lines with conventional low-frequency lines 4 5 10

HOT 3/24 69 Bh, 71 Ma, 79 Ch · builds on §1.2; here the focus is the *line* rather than the circuit

A transmission line is a guiding structure

- At low frequency, power is thought of as travelling **in the wire**; the pair of conductors merely completes a circuit.
- At microwave frequency, power travels **in the $\vec{E}$ and $\vec{H}$ fields** between and around the conductors. Any structure that guides that field from place to place is a transmission line.
- The governing criterion is again d/λ (§1.2): once the line length is a noticeable fraction of λ , voltage and current are functions of position.

Classification

- Two-conductor (TEM) lines** — support a pure TEM mode, no low-frequency cut-off:
 - Two-wire open line, twisted pair (UTP / STP), coaxial cable, stripline.
- Quasi-TEM lines** — inhomogeneous dielectric, so no exact TEM:
 - Microstrip, coplanar waveguide (CPW), slot line, fin line.
- Single-conductor (waveguide) lines** — TE/TM modes only, and a **cut-off frequency** below which nothing propagates:
 - Rectangular and circular waveguide, dielectric waveguide, optical fibre.
- Balanced vs unbalanced**: two-wire and twisted pair are balanced (equal and opposite currents, common-mode noise cancels); coaxial is unbalanced (signal referred to the grounded outer). A **balun** converts between them.

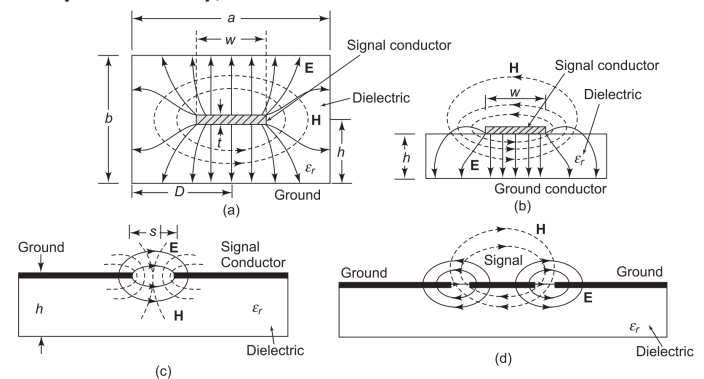
Choosing a line — the four practical trade-offs

- Loss**: waveguide < coax < microstrip at a given frequency.
- Power handling**: waveguide (kW–MW) $\gg$ coax $\gg$ microstrip (W).
- Bandwidth**: TEM lines are broadband; waveguide is limited by cut-off and the next higher mode.
- Cost and integration**: microstrip is photolithographic and near-free; waveguide is machined and expensive.

Low-frequency line vs microwave line

Aspect	Conventional (LF)	RF / Microwave
Model	Ideal short; length irrelevant.	Distributed R', L', G', C' per metre.
Analysis	KVL / KCL, lumped.	Telegrapher's equations, Maxwell.
V, I	Uniform along the line.	$v(z, t)$, $i(z, t)$; standing waves.
Loss	Ohmic ($I^2 R$) only.	Conductor ($\propto \sqrt{f}$, skin effect), dielectric ($\tan \delta$), and radiation.
Reflection	None considered.	At every discontinuity; measured by Γ , VSWR.
Typical lines	Wire, twisted pair, PCB trace.	Coax, waveguide, microstrip, stripline.
Termination	Matching not needed.	Must terminate in Z_0 or match with a tuner.
Joints	Any solder joint.	Precision connectors (SMA, N, APC-7); a bad joint is a reflector.

The planar family, in cross-section



(a) stripline, (b) microstrip, (c) slot line, (d) coplanar strips, with E (solid) and H (dashed). Only (a) has one homogeneous dielectric round the conductor, so only (a) is pure TEM. Das & Das Fig 3.27.

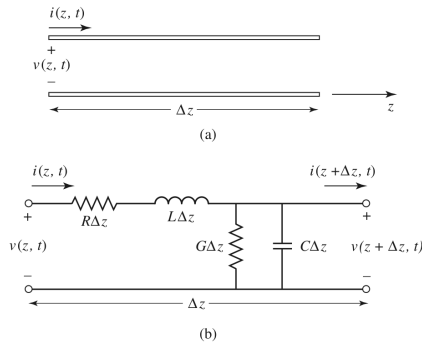
→ For microstrip vs stripline (asked 71 Ma), see §1.6 — geometry, quasi-TEM vs pure TEM, dispersion, shielding and fabrication are compared there and are not repeated.

2.2 Transmission Line Theory

Derive the transmission line (telegrapher's) equations and obtain the characteristic impedance, reflection coefficient, VSWR and input impedance 4 6 8

HOT 3/24 69 Bh, 79 Ch, 71 Ma · the toolkit every numerical in this chapter uses

Distributed model and the telegrapher's equations



(a) v, i on a length Δz ; (b) its equivalent cell. Draw (b) before writing KVL and KCL.

Pozar Fig 2.1.

- A length Δz of line is modelled by series $R'\Delta z, L'\Delta z$ and shunt $G'\Delta z, C'\Delta z$, all **per unit length**.
- Applying KVL and KCL to that cell and letting $\Delta z \rightarrow 0$:

$$\frac{\partial v}{\partial z} = -R'i - L'\frac{\partial i}{\partial t}, \quad \frac{\partial i}{\partial z} = -G'v - C'\frac{\partial v}{\partial t}$$

- In phasor form these decouple into the **wave equations**

$$\frac{d^2 V}{dz^2} = \gamma^2 V, \quad \frac{d^2 I}{dz^2} = \gamma^2 I$$

- **Propagation constant and characteristic impedance:**

$$\gamma = \alpha + j\beta = \sqrt{(R' + j\omega L')(G' + j\omega C')}$$

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}}$$

- Solution: $V(z) = V_0^+ e^{-\gamma z} + V_0^- e^{+\gamma z}$ – a forward and a reflected travelling wave.

Lossless line ($R' = G' = 0$) – used for every PYQ in this chapter

- $\alpha = 0, \quad \beta = \omega\sqrt{L'C'}, \quad Z_0 = \sqrt{L'/C'}$ (purely real).
- $\lambda = 2\pi/\beta = v_p/f, \quad v_p = 1/\sqrt{L'C'} = c/\sqrt{\epsilon_r}$.

Distortionless line

- Condition: $R'C' = L'G'$. Then $\alpha = \sqrt{R'G'}$ is frequency-independent and $\beta = \omega\sqrt{L'C'}$ is linear in ω , so all frequencies travel at the same v_p and the pulse shape survives.

Terminated line: the four quantities every problem needs

- **Load reflection coefficient**

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{z_L - 1}{z_L + 1}, \quad z_L = \frac{Z_L}{Z_0}$$

- **At a distance d toward the generator**, only the phase changes on a lossless line:

$$\Gamma(d) = \Gamma_L e^{-j2\beta d}, \quad |\Gamma(d)| = |\Gamma_L|$$

This is why the Smith chart works: moving along the line is a *rotation at constant radius*.

- **VSWR**

$$S = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}, \quad 1 \leq S \leq \infty$$

- **Input impedance**

$$Z_{in}(d) = Z_0 \frac{Z_L + jZ_0 \tan \beta d}{Z_0 + jZ_L \tan \beta d}$$

- **Return loss** = $-20 \log_{10} |\Gamma|$ dB; **mismatch loss** = $-10 \log_{10} (1 - |\Gamma|^2)$ dB.

Special line lengths – memorise these

Case	Input impedance
Matched $Z_L = Z_0$	$Z_{in} = Z_0$ for any length; $\Gamma = 0, S = 1$.
Short $Z_L = 0$	$Z_{in} = jZ_0 \tan \beta d$ – purely reactive, inductive for $d < \lambda/4$.
Open $Z_L = \infty$	$Z_{in} = -jZ_0 \cot \beta d$ – purely reactive, capacitive for $d < \lambda/4$.
$d = \lambda/2$	$Z_{in} = Z_L$ – the line repeats every half wavelength.
$d = \lambda/4$	$Z_{in} = Z_0^2/Z_L$ – the quarter-wave transformer ; it inverts , so a short looks open.
$d = \lambda/8$	$Z_{in} = Z_0 \frac{Z_L + jZ_0}{Z_0 + jZ_L}$ ($\tan \beta d = 1$).

A shorted or open stub is therefore a *pure reactance whose value you choose by cutting it to length* – that single fact is the whole basis of stub matching.

2.3 The Smith Chart

Explain the construction of the Smith chart. What does each family of curves represent, and how is it read? 4 5 10

HOT 5/24 72 Ash, 75 Bh, 72 Ma, 81 Ba, 80 Ch · counted together with the immittance chart in §2.4

What the chart is

- A plot of **normalised impedance** $z = r + jx$ mapped onto the complex Γ plane by the bilinear transform

$$\Gamma = \frac{z - 1}{z + 1}, \quad z = \frac{1 + \Gamma}{1 - \Gamma}$$

- Because $|\Gamma| \leq 1$ for a passive load, the entire right half of the z plane folds into the **unit circle** – an infinite plane on one sheet of paper.

The two families of curves

- Constant resistance r :** circles centred on $\left(\frac{r}{1+r}, 0\right)$ with radius $\frac{1}{1+r}$. All pass through the open-circuit point $\Gamma = +1$.
- Constant reactance x :** arcs centred on $\left(1, \frac{1}{x}\right)$ with radius $\left|\frac{1}{x}\right|$. Upper half = $+x$ (**inductive**), lower half = $-x$ (**capacitive**).
- The two families are **orthogonal** everywhere they cross.

Landmarks

- Centre** $z = 1$: matched, $\Gamma = 0$, $S = 1$.
- Left** $z = 0$: short circuit, $\Gamma = -1$.
- Right** $z = \infty$: open circuit, $\Gamma = +1$.
- Rim** $|\Gamma| = 1$: pure reactance, total reflection, $S = \infty$.
- Horizontal axis: $x = 0$, i.e. purely resistive. To the right of centre lies $V_{\max}$ (and $r = S$); to the left lies $V_{\min}$ (and $r = 1/S$).

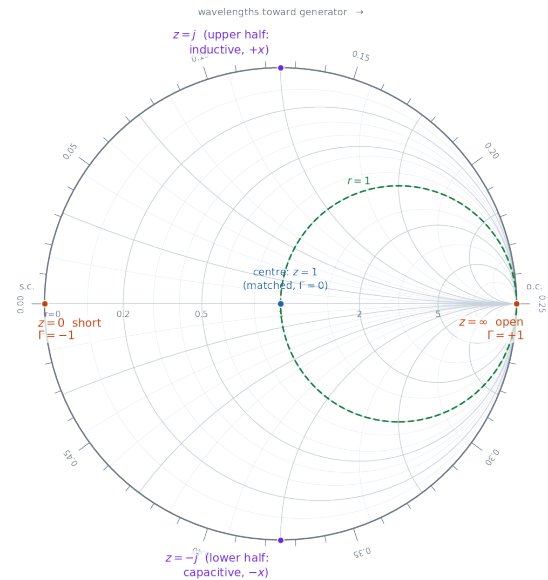
Reading it – the five rules

- Normalise: $z_L = Z_L/Z_0$. Plot it.
- $|\Gamma_L|$ = distance from centre $\div$ chart radius; its angle is read on the **angle of reflection coefficient** scale.
- Draw the **constant- $|\Gamma|$ (SWR) circle** through the point. Read S where it cuts the right-hand real axis.
- Move toward the generator = clockwise**; toward the load = anticlockwise. **One full lap is $\lambda/2$** , so a quarter turn is $\lambda/8$.
- Read distances on the **WTG** (wavelengths toward generator) rim scale, which runs $0 \rightarrow 0.5$ clockwise starting at the short-circuit point.

Why the chart is worth using at all

- It turns the awkward Z_{in} formula into a **rotation**, and $z \rightarrow y$ into a **half-turn** – both done with a compass instead of complex arithmetic.
- Γ , S , Z_{in} , Y_{in} , return loss and the positions of $V_{\max}$ and $V_{\min}$ are all read off one diagram.
- It shows *graphically* which matching solutions exist, and lets you pick the one with the shortest stub or the widest bandwidth – something a formula does not show you.

The Smith chart: constant- r circles and constant- x arcs



every point is one normalised impedance $z = r + jx$; $|\Gamma|$ is its distance from the centre, so one lap of the rim is $\lambda/2$ of line

The impedance chart. Constant- r circles (all through the open point) and constant- x arcs; the rim carries the WTG scale, on which one complete lap is half a wavelength of line.

2.4 The Immittance (Admittance) Chart

Sketch an immittance chart. List the duality parameters and compare the two scales 2 4 5 10

HOT 5/24 72 Ash, 75 Bh, 72 Ma, 81 Ba, 80 Ch · short, self-contained, and asked five times

The admittance chart

- Since $y = 1/z$, and

$$\Gamma_y = \frac{y - 1}{y + 1} = -\frac{z - 1}{z + 1} = -\Gamma_z$$

the admittance chart is the **impedance chart rotated through 180°** (a $\lambda/4$ line length).

- So on a plain Smith chart you convert $z \rightarrow y$ by drawing a diameter through the point and reading the diametrically opposite intersection.

The immittance chart

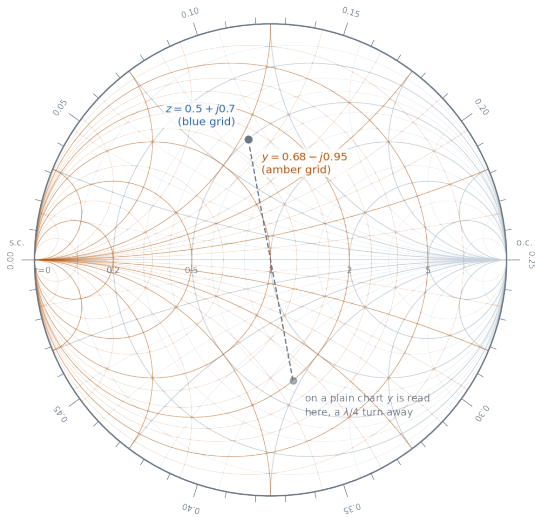
- Both grids printed on one chart, usually in two colours: the impedance grid and the admittance grid superimposed.
- A single location then carries **two readings at once** — z on one grid, y on the other. The $\lambda/4$ half-turn is built into the printing.
- Why it matters:** stub matching is done in *admittance* (shunt stubs add susceptance) but loads are quoted as *impedance*. Without an immittance chart every problem costs an extra half-turn, and each half-turn is a chance to misread.

Duality parameters

Impedance grid	Admittance grid
$z = r + jx$	$y = g + jb$
Resistance r	Conductance g
Reactance x	Susceptance b
$+x$ inductive (upper)	$+b$ capacitive (upper)
$-x$ capacitive (lower)	$-b$ inductive (lower)
Short $z = 0$ at $\Gamma = -1$	Open $y = 0$ at $\Gamma = -1$
Open $z = \infty$ at $\Gamma = +1$	Short $y = \infty$ at $\Gamma = +1$
Series elements add	Shunt elements add
$r = 1$ circle	$g = 1$ circle

Trap. The sign convention flips: a point in the *upper* half is inductive read as an impedance but *capacitive* read as an admittance. Both are the same physical point.

Immittance chart = impedance grid + admittance grid
wavelengths toward generator $\rightarrow$



one location, two readings: the $\lambda/4$ half-turn that converts $z \rightarrow y$ is printed into the second grid, so no rotation is needed

Immittance chart. The amber grid is the blue grid turned through 180° ; one location reads $z = 0.5 + j0.7$ on the impedance grid and $y = 0.68 - j0.95$ on the admittance grid. The faint point is where y would have to be read on a plain chart — a $\lambda/4$ turn away.

2.5 Impedance Matching: Why, and Which Technique

Why is impedance matching necessary? Compare the different impedance matching techniques 4 5

6

HOT 3/24 78 Ch, 74 Ma, 77 Ch · the theory half of several double-stub questions

Why match

- **Maximum power transfer:** a matched load absorbs all incident power; the fraction reflected is $|\Gamma|^2$.
- **Protect the source:** reflected power returning to a high-power amplifier or a magnetron causes heating, and can destroy it.
- **Remove standing waves:** high VSWR means voltage peaks that cause dielectric breakdown and arcing, and current peaks that cause I^2R loss.
- **Preserve receiver sensitivity:** mismatch at an LNA input degrades the noise figure directly.
- **Avoid distortion:** multiple reflections on a long line arrive delayed, producing ghosting and intersymbol interference.

The matching network must be lossless

- A resistive pad would “match” but dissipate the very power being matched.
- So matching uses only **reactive** elements: line lengths, stubs, lumped L and C .



The general problem: a lossless network between line Z_0 and load Z_L , chosen so the line sees Z_0 . Every technique in the table is one way to build the box. Pozar Fig 5.1.

Comparison of techniques

Technique	How it works	Trade-offs
Quarter-wave transformer	$\lambda/4$ of line with $Z_{01} = \sqrt{Z_0 Z_L}$.	Simplest, but needs a real Z_L – a complex load must first be walked to a $V_{\max}$ or $V_{\min}$. Narrow-band.
L-section (lumped)	Two reactances, series + shunt.	Compact; fine below ~ 1 GHz. Above that, component parasitics (\$1.3) dominate.
Single stub	One shunt (or series) stub at a computed distance d .	Matches any load; only line, so no lumped parts. But d must move if the load changes – awkward to build as an adjustable tuner.
Double stub	Two stubs at fixed spacing; only the lengths are tuned.	Positions are preset, so it makes a practical adjustable tuner. But a range of loads cannot be matched (the forbidden region).
Triple stub	Three fixed stubs.	Removes the forbidden region entirely; used in laboratory slide-screw tuners.

Advantages of double stub over single stub

- Stub **positions are fixed**; matching is done by adjusting lengths only, so it can be built as a mechanical tuner with two sliding shorts.
- No need to break into the line at a computed distance – the connection points are decided at manufacture.
- Retunes for a new load without physically relocating anything.
- **Cost:** a set of load admittances (the forbidden region) is unmatchable, and two stubs mean a slightly narrower bandwidth.

2.6 Single-Stub Tuning

Explain the design of a single-stub shunt tuner, stating all steps with reasoning. When is a short-circuited stub preferred over an open one? 8 10 2+8

HOT 6/24 72 Ash, 74 Bh, 72 Ma, 76 Bh, 79 Ch, 81 Bh · worked instances in the numerical companion

Principle

- Two degrees of freedom are needed to cancel a complex mismatch, and a single stub supplies exactly two: **the distance d from the load to the stub**, and **the stub length ℓ** .
- Work in **admittance**, because a shunt stub *adds* its susceptance:

$$y_{in} = y(d) + y_{stub} = 1 + jb + (-jb) = 1$$

- Step 1 — choose d** so the line admittance has unit conductance: $y(d) = 1 + jb$. Geometrically, walk round the SWR circle until it cuts the $g = 1$ circle. It always cuts it **twice**, so there are always two solutions.
- Step 2 — choose ℓ** so the stub presents $-jb$, cancelling the residual susceptance and leaving $y_{in} = 1$, i.e. $Z_{in} = Z_0$.

Design procedure on the chart

- Normalise $z_L = Z_L/Z_0$ and plot it.
- Half-turn to y_L (or read it on the admittance grid). Note its WTG reading.
- Draw the SWR circle through y_L .
- Move **clockwise** to each intersection with the $g = 1$ circle; call them $y_1 = 1 - jb$ and $y_2 = 1 + jb$. The WTG difference from y_L gives d_1 and d_2 .
- The stub must supply $+jb$ and $-jb$ respectively.
- For the stub length, start at the **termination point of the stub** — the open-circuit point ($y = 0$) for an open stub, the short-circuit point ($y = \infty$) for a shorted stub — and run **clockwise round the rim** to the required susceptance. That arc, read on the WTG scale, is ℓ .

Analytic check (use it to verify the chart)

- With $t = \tan \beta d$ and $Z_L = R_L + jX_L$, the distance satisfies

$$t = \frac{X_L \pm \sqrt{R_L[(Z_0 - R_L)^2 + X_L^2]}/Z_0}{R_L - Z_0}$$

reducing to $t = -X_L/2Z_0$ when $R_L = Z_0$.

- Then $d/\lambda = \frac{1}{2\pi} \tan^{-1} t$ (add 0.5 if negative).
- Stub length from the required susceptance b_s :**

$$\frac{\ell_{open}}{\lambda} = \frac{1}{2\pi} \tan^{-1}(b_s), \quad \frac{\ell_{short}}{\lambda} = \frac{-1}{2\pi} \tan^{-1}\left(\frac{1}{b_s}\right)$$

always reduced modulo 0.5λ .

Short-circuited or open-circuited stub?

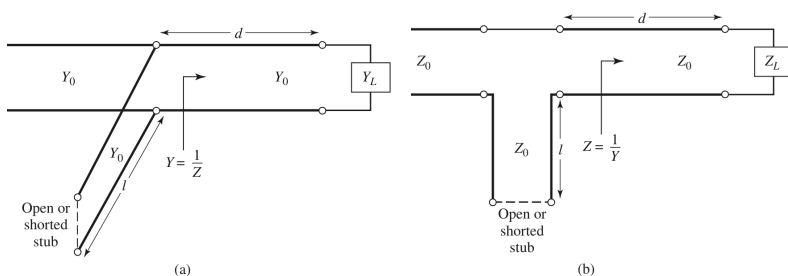
Stub	When it is preferred
Short	Coax and waveguide. A shorting plate is exact and radiates nothing; an open coax end radiates and has fringing capacitance, so its electrical length is not its physical length. Also easier to make adjustable (sliding short).
Open	Microstrip and other planar lines. Ending the strip is free; a true short needs a via drilled and plated through the substrate, which adds inductance and cost.

The two lengths differ by exactly $\lambda/4$: $\ell_{short} = \ell_{open} \pm \lambda/4$.

Which of the two solutions to build

- Prefer the **smaller d** : less line between load and stub means a shorter high-VSWR section, hence lower loss and **wider bandwidth**.
- Prefer the **shorter stub** for the same reason, and for compactness.
- A series stub is used when the line cannot be shunted (some waveguide and coaxial geometries); the construction is identical but done on the *impedance* grid against the $r = 1$ circle.

The circuit, which is what to draw



- (a) Shunt stub:** at distance d from the load the line admittance is $Y = 1/Z = Y_0 + jb$; the stub of length ℓ , open or shorted, adds $-jb$.
- (b) Series stub:** same two unknowns d and ℓ , but the stub is in series, so work in impedance: $Z = Z_0 + jX$ at d , stub supplies $-jX$.
- Mark d from the **load** and ℓ from the stub's **open or shorted end** on the sketch: those are the two answers the question asks for.
- Pozar Fig 5.4.

2.7 Double-Stub Tuning

Explain the steps for impedance matching of a load using a double-stub network. State the advantages over single-stub matching and the limitation of the technique **8 10 12 2+8**

TOP 17/24 69 Bh, 70 Bh, 70 Ma, 71 Bh, 73 Bh, 73 Ma, 74 Ma, 75 Bh, 77 Ch, 78 Ch, 80 Ch, 80 Ba, 80 Bh, 81 Ba, 79 Bh, 82 Ba, 82 Bh · **the single most asked item in the subject**

Principle

- The two stubs sit at **fixed** positions a distance d apart – almost always $\lambda/8$ or $3\lambda/8$ in the papers. Only the two lengths ℓ_1 and ℓ_2 are adjusted, and those are the two degrees of freedom.
- **Stub 1** cannot change the conductance it sees – a shunt susceptance moves the point *along its own constant- g circle*. Its job is to land the point where the intervening d of line will carry it onto the $g = 1$ circle.
- **Stub 2** then simply cancels whatever susceptance is left, exactly as in single-stub matching.

The spacing circle – the key construction

- Define the **spacing circle** as the $g = 1$ circle rotated d **toward the load** (anticlockwise).
- Any point on it is carried *onto* $g = 1$ by d wavelengths of line.
- So stub 1 must move the point from y_1 to the intersection of its own constant- g circle with the spacing circle. There are generally **two** intersections, hence two solutions.

Design procedure

1. Normalise z_L , plot, half-turn to y_L .
2. If the first stub is offset from the load by d_1 , rotate y_L clockwise by d_1 to get y_1 , the admittance *just before* stub 1.
3. Draw the spacing circle ($g = 1$ rotated d toward the load).
4. Follow the **constant- g circle through** y_1 to its intersections with the spacing circle: y_a and y'_a .
5. $b_1 = \text{Im}(y_a) - \text{Im}(y_1)$ – the susceptance stub 1 must add. Read ℓ_1 off the rim from the stub's termination point.
6. Rotate y_a clockwise by d along its SWR circle; it lands on $g = 1$ at $y_2 = 1 + jb'_2$.
7. $b_2 = -b'_2$; read ℓ_2 off the rim the same way. The network is matched.

Analytic check

- With $y_1 = g_1 + jb_L$ and $t = \tan \beta d$, requiring $\text{Re}(y_2) = 1$ after the spacing gives

$$\text{Re}(y_2) = \frac{g_1(1+t^2)}{(1-tB)^2 + t^2g_1^2} = 1$$

$$\Rightarrow B = \frac{1 \pm \sqrt{g_1(1+t^2) - g_1^2t^2}}{t}, \quad b_1 = B - b_L$$

where B is the total susceptance after stub 1. **Note the leading 1 in the numerator, not g_1 – a common slip.**

- Then $y_a = g_1 + jB$, and $y_2 = \frac{y_a + jt}{1 + jt y_a}$, with $b_2 = -\text{Im}(y_2)$.

The forbidden region – always state this

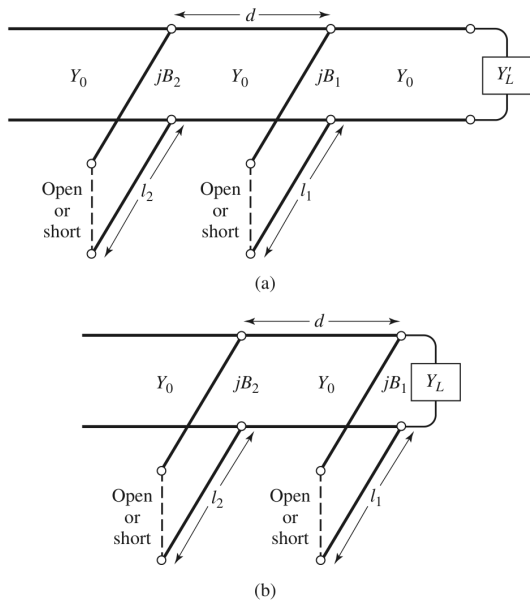
- The square root is real only if

$$g_1 \leq \frac{1}{\sin^2 \beta d}$$

- For $d = \lambda/8$ ($\beta d = 45^\circ$): $g_1 \leq 2$.
For $d = 3\lambda/8$: $g_1 \leq 2$ likewise.
For $d = \lambda/4$: $g_1 \leq 1$.
- Loads whose conductance at the first stub exceeds this **cannot be matched** at all with that spacing. Graphically, their constant- g circle lies wholly inside the spacing circle and never cuts it.
- **The fix:** add a length of line before the first stub (which moves g_1), or use a third stub.
- This is why $\lambda/4$ spacing is avoided in practice and $\lambda/8$ or $3\lambda/8$ is used – and why the examiner keeps specifying $3\lambda/8$.

Trap. Stub spacings near $\lambda/2$ make the two stubs nearly coincident and the tuner hopelessly narrowband, so practical spacings stay between $\lambda/8$ and $3\lambda/8$.

The circuit, which is what to draw



- (a) load an arbitrary distance from the first stub: stub 1 (jB_1 , length ℓ_1) nearer the load, stub 2 (jB_2 , ℓ_2) a fixed d further toward the generator.
- (b) the same tuner with that offset line absorbed into the load, now called Y_L' . This is step 2 of the procedure: rotate y_L by d_1 first, then design as if stub 1 sat at the load.
- Only ℓ_1 and ℓ_2 are adjusted; d is fixed at manufacture, which is the whole advantage over one stub.
- Pozar Fig 5.7.

2.8 S-Matrix of a Matched Network

Prepare the scattering matrix of the network before and after matching 2 2+2 4 6

TOP 10/24 76 Bh, 77 Ch, 78 Ch, 80 Ba, 80 Bh, 81 Ba, 81 Bh, 79 Bh, 82 Ba, 82 Bh · free marks: two lines of work, tacked onto ten stub questions

Before matching — a one-port

- An unmatched load seen from the line is a **one-port**, so its scattering matrix is the single entry

$$[S] = [\Gamma_L] = \left[\frac{Z_L - Z_0}{Z_L + Z_0} \right]$$

- Quote it in polar form, $|\Gamma_L| \angle \theta^\circ$, and give the VSWR alongside — the examiner usually wants both.

After matching — an ideal two-port

- A perfectly matched, lossless, reciprocal two-port has **no reflection at either port**, so $S_{11} = S_{22} = 0$, and all power passes through, so $|S_{12}| = |S_{21}| = 1$:

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Keeping the electrical length $\theta = \beta \ell$ of the through path:

$$[S] = \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

Either answer earns the marks; state which convention you are using.

Justifying each property (this is where the marks are)

- $S_{11} = S_{22} = 0$: that is the definition of a match — the tuner was designed to make $Z_{in} = Z_0$ looking in at either port.
- **Reciprocity**, $S_{12} = S_{21}$: the network contains only passive, isotropic media (line and stubs), no ferrite and no active device, so $[S]$ is symmetric, $[S] = [S]^T$.
- **Lossless**, so $[S]$ is **unitary**: $[S]^{*T}[S] = [I]$, giving $\sum_k |S_{ki}|^2 = 1$ for every column. With $S_{11} = 0$ this forces $|S_{21}| = 1$.
- **Reactive elements only** means no power is dissipated, only redirected — so all incident power appears at the far port.

Worked shape of the answer

- “Before: $\Gamma_L = (Z_L - Z_0)/(Z_L + Z_0) = 0.45 \angle 60^\circ$, so $[S] = [0.45 \angle 60^\circ]$ and $S = 2.64$.”
- “After: the tuner makes $Z_{in} = Z_0$, so $S_{11} = S_{22} = 0$; the network is reciprocal and lossless, so $S_{12} = S_{21}$ with $|S_{21}| = 1$, giving $[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.”

S-parameter theory in full — properties, derivation, shifting reference planes — is Chapter 3. This band covers only the rider attached to stub questions.

verified/added The 3-port and 4-port version in class notes

- Some class notes draw each stub as an extra port: single stub $\rightarrow 3 \times 3$, double stub $\rightarrow 4 \times 4$, then write “after perfect matching” as a matrix whose only non-zero entry is S_{21} (or $-S_{21}$ for a 180° output), and “power divided equally” as $S_{11}/3$, $S_{21}/3$, $S_{31}/3$ down one column.
- Do not copy those matrices.** Each fails a test of §3.3:
 - $S_{21} \neq 0$ with $S_{12} = 0$ is **not symmetric**, but line and stubs are passive and isotropic, so the network must be reciprocal.
 - One column of $S_{k1}/3$ does not sum to 1 in power, and the other columns are zero, so the matrix is **not unitary** for a lossless tuner.

verified/added What to write if the stub ports are wanted

- A stub's far end is a **short or open**, a fixed reactive termination with $|\Gamma| = 1$. It is not an external port, so it is absorbed into the network and the tuner seen from the line is the **two-port** above.
- Sketch the stubs as ports only to label the junctions, then reduce: before matching $[\Gamma_L]$, after matching $\begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$.
- Terminating the stub arm in Z_0 instead turns the tuner into a plain tee junction, and by the 3-port theorem (§3.5) a tee cannot be lossless, reciprocal and matched at every port at once. That one line earns the extra marks.

2.9 Microstrip Realisation of a Stub Tuner**Sketch the physical diagram of the matched network using microstrips** 2 2+2 4**HOT 4/24** 72 Ash, 72 Ma, 77 Ch, 82 Ba · turns λ answers into millimetres**From electrical length to millimetres**

- Microstrip is **quasi-TEM**, so the wave sees an effective permittivity between air and substrate:

$$\epsilon_e = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}}$$

- Guide wavelength, and hence every physical length:

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_e}} = \frac{c}{f\sqrt{\epsilon_e}}, \quad \ell_{\text{mm}} = \frac{\ell}{\lambda} \times \lambda_g$$

- Strip width** for a wanted Z_0 (Hammerstad synthesis), with $B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}$ and $W/h > 2$:

$$\frac{W}{h} = \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left(\ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right]$$

Layout rules the examiner looks for

- Stubs are drawn as **T-junctions off the main line**, at the computed spacing.
- Open stubs** are drawn as a plain truncated strip; **shorted stubs** need a **via to the ground plane**, drawn as a filled circle.
- Label W for the Z_0 line, the stub lengths ℓ_1 , ℓ_2 in mm, and the spacing d in mm.
- State the substrate: e.g. **RT/duroid 5880** ($\epsilon_r = 2.2$), **FR-4** ($\epsilon_r = 4.4$), **alumina** ($\epsilon_r = 9.8$), with h .

Worked conversion — carry this shape into the exam

- Take $Z_0 = 50 \Omega$ on FR-4, $\epsilon_r = 4.4$, $h = 1.6$ mm, $f = 3$ GHz.
- Synthesis gives $W/h \approx 1.9$, so $W \approx 3.0$ mm, and $\epsilon_e \approx 3.33$.
- $\lambda_0 = c/f = 100$ mm, so $\lambda_g = 100/\sqrt{3.33} \approx 54.8$ mm.
- A stub of 0.146λ is then $0.146 \times 54.8 \approx \mathbf{8.0}$ mm, and a $3\lambda/8$ spacing is $0.375 \times 54.8 \approx \mathbf{20.5}$ mm.

Practical corrections worth a mark

- Open-end fringing:** an open stub behaves slightly longer than it is, by $\Delta\ell \approx 0.3\text{--}0.4h$. Shorten the drawn stub by that much.
- T-junction discontinuity:** the junction adds shunt susceptance; the reference plane is not exactly the strip centreline.
- Via inductance:** a shorted stub's via is not a perfect short, which is the practical reason open stubs dominate in microstrip.
- Radiation and dispersion** rise with frequency; above about 10 GHz on FR-4 the losses make the design impractical, so low-loss substrates are used.

2.10 PYQ Mapping

Theory questions – covered in this document

- Types of transmission lines; LF line vs microwave line – §2.1. 69 Bh 4, 79 Ch 10
- Microstrip vs stripline – §1.6. 71 Ma 5
- Smith chart construction and reading – §2.3.
- Immittance / admittance chart, duality, scale comparison – §2.4. 72 Ash 2, 75 Bh 10, 72 Ma 5, 81 Ba 4, 80 Ch 5
- Impedance matching techniques compared; advantages of double over single stub – §2.5. 78 Ch 4+6, 74 Ma 2+8, 77 Ch 2+8
- Single-stub design steps and reasoning – §2.6. 79 Ch 10, 81 Bh 5+3+2, 76 Bh 10
- Double-stub design steps; forbidden region – §2.7. 74 Ma 2+8, 78 Ch 4+6
- S-matrix before and after matching – §2.8. 10 papers, see the band
- Microstrip realisation of the matched network – §2.9. 72 Ash, 72 Ma, 77 Ch, 82 Ba

Numerical questions – solved in the numerical companion

- Smith chart reading (Γ_L , VSWR, Z_{in} , first resistive point): 79 Ch 10 – Problem 1.
- Single stub (6): 72 Ash, 74 Bh, 72 Ma, 76 Bh, 79 Ch, 81 Bh – Problems 2–7.
- Double stub (17): 69 Bh, 70 Bh, 70 Ma, 71 Bh, 73 Bh, 73 Ma, 74 Ma, 75 Bh, 77 Ch, 78 Ch, 80 Ch, 80 Ba, 80 Bh, 81 Ba, 79 Bh, 82 Ba, 82 Bh – Problems 8–24.

Syllabus coverage check

- Types of transmission lines – §2.1 ✓
- Transmission line theory – §2.2 ✓
- Smith Chart analysis – §2.3, §2.4 ✓
- Impedance transformations and matching analysis – §2.5–§2.9 and the numerical companion ✓

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.